File System and Tables Exercises

1. What is a major advantage of organizing information into tables?

A. Tables eliminate the need to update information.
B. Tables make information easier to search, compare, and summarize.
C. Tables prevent all unauthorized access.
D. Tables automatically remove duplicate information.

2. What do rows and columns represent in a table?

A. Rows represent records; columns represent attributes.
B. Rows represent attributes; columns represent records.
C. Rows represent files; columns represent databases.
D. Rows represent databases; columns represent relationships.

3. Why is a CustomerID field useful?

A. It stores all information about a customer.
B. It prevents customers from changing their names.
C. It eliminates the need for other attributes.
D. It distinguishes customers who might have the same name.

4. What is data redundancy?

A. Related data being difficult to combine
B. Unauthorized users accessing confidential information
C. The same information being stored in multiple places
D. Multiple transactions updating data simultaneously

5. A customer's address is updated in one file but remains unchanged in another. What problem does this illustrate?

A. Data isolation
B. Data inconsistency
C. A concurrency problem
D. A security problem

6. Which design reduces unnecessary repetition of department informations?

A. Store department info in a Department table and reference them using DepartmentID.
B. Store the department name in every employee record.
C. Create a separate file for every employee.
D. Remove all department information.

7. What is data isolation?

A. Storing the same information in multiple places
B. Restricting access to confidential information
C. Two users updating the same record simultaneously
D. Difficulty accessing and combining related data scattered across separate files or systems

8. Related files store dates in different formats. Why can this be a problem?

A. Different date formats always represent different dates.
B. Dates cannot be stored in files.
C. Applications may need to convert the formats before combining and processing the data.
D. Different formats automatically delete duplicate records.

9. A bank teller can view confidential loan information they are not authorized to access. What problem does this illustrate?

A. Data redundancy
B. A security problem
C. Data isolation
D. A concurrency problem

10. Two transactions read the same account balance, then one overwrites the other's update. What problem does this illustrate?

A. A concurrency problem
B. Data isolation
C. A file-format problem
D. A security problem